ALA: 1-Python

AIM: To write a Python program that generates the terms of a Geometric Progression using a while loop, checks whether each term is even or odd, calculates the total sum of the terms, and determines whether the series is constant or changing.

Error code:

```
print("Factor Finder")
number = int(input("Enter number: "))
i = 1
count = 0
sum = 0

while i <= number:

if number % i == 0:

print("Factor:", i)
count = count + 1
sum = sum + i
i = i + 1
print("Total factors:", count)
print("Sum of factors:", sum)

if sum > number:
print("Abundant number")

else if sum == number:
print("Perfect number")

else:
print("Deficient number")
```

Errors:

- **Error 1: Indentation Error**
- is not indented properly.
- Python **requires indentation** inside loops and conditions.
- Correct: if number % i == 0:

- Error 2: Wrong Syntax → else if

- In Python, else if is wrong.
- The correct keyword is: elif

➢ Error 3: Logic Mistake (Sum Includes the Number Itself)
- We compare using **sum of proper factors** (excluding the number itself).
- Correct: sum = sum – number

➢ Error 4: Indentation error in if
- If number % I == 0:
  Print(“factor:” , i)
  Count = count + 1
- Correct:
- If number % I == 0:
  Print(“factor:” , i)
  Count = count + 1

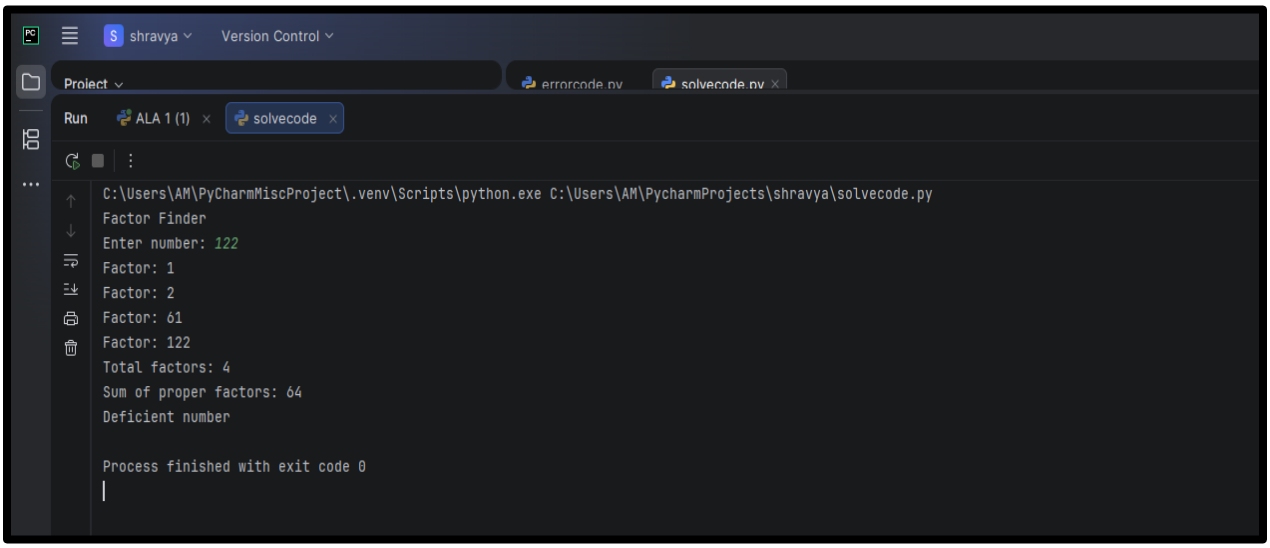
- Error 5: Indentation error in if
- if sum > number:
  print("Abundant number")
- correct: if sum > number:
  print("Abundant number")

- Error 6: Indentation error else if
- else if sum == number:
  print("Perfect number")
- correct: elifsum == number:
  print("Perfect number")

- Error 7: Indentation error else
- else:
  print("Deficient number")
- correct: else:
  print("Deficient number")

SOLUTION:

```
print("Factor Finder")
number = int(input("Enter number: "))
i = 1
count = 0
total_sum = 0

while i <= number:
    if number % i == 0:
```

```
        print("Factor:", i)
        count = count + 1
        total_sum = total_sum + i
    i = i + 1

print("Total factors:", count)

# Subtract the number itself to get proper factor sum
proper_sum = total_sum - number

print("Sum of proper factors:", proper_sum)

if proper_sum > number:
    print("Abundant number")
elif proper_sum == number:
    print("Perfect number")
else:
    print("Deficient number")
```

OUTPUT: